1. 01-endo-chain: factor bases closed under the degree-5 CM endomorphism (pair 46)

2026-10-15. Pair-46 factor-base campaign, item 01. Protocol `PROTOCOL.md` (sha256 0ce98a09...dfcf2), frozen before data. Instrument: standalone Rust binary `src/main.rs`, 1 core, L0 timing. All 1 600 phase-1 solves and 800 post-hoc solves were verified by d·P = Q.

Verdict: negative, with an exact reason (E3). The best endo-chain arm, a single chain from P, costs $S = 1.47\,[1.28, 1.69]$ group-op equivalents per $\sqrt{N}$ on the 22-bit toy. Strided ±BSGS costs $0.955\,[0.86, 1.05]$ on the same targets (ratio $1.54\,[1.30, 1.82]$), and the rho-with-negation walk takes $0.88\sqrt{N}$ steps. Chains do beat the abscissa (small-x) base, by a factor of about $\sqrt{k}$ at fixed $|\mathrm{FB}|$. That is a constant, and it ends at the single-chain search, which stays above BSGS.

1.1. Setting

Pair 46: $p = 5947698030651343$, $N = 5947697996161643$, $p = 5t^2 + 19t + 19$, $t^2 - 4p = -19(t+2)^2$. The endomorphism $w = \frac{1+\sqrt{-19}}{2}$ has degree 5 and eigenvalue $\lambda = 2973849084305079$. Its x-map $R = \frac{\mathrm{Num}_5}{\mathrm{Den}_4}$ was interpolated and verified on 1000 points, and $\mathrm{ord}(\lambda) = \frac{N-1}{7} = 849671142308806$. Toy curves come from the same family: the crater curve $j = -96^3$ with prime order at 16, 18, 20 and 22 bits.

1.2. Exact statements

- **E2.** There is one unknown per chain: $\log \psi^i H = \lambda^i \log H$. Two hits on the same chain give a linear equation in d.
- **E3.** A known-head chain with an x-only table is BSGS with table entries costing c_E each and unstrided giant steps. Its optimum is $\sqrt{2c_E N}$ against $\sqrt{N}$ for strided ±BSGS. More chains with unknown heads add birthday waiting, so one chain is optimal.
- **E4.** $\mathrm{Aut} = \{\pm 1\}$, so glv-orbit reduces to the ± fold. Folding the walk by ψ-orbits needs canonical representatives of orbits of length $8.5 \cdot 10^{14}$, which is infeasible.
- **E5.** Phase-1 artefact: with the fixed-b walk $Q + aP$, the two hit positions of an element are symmetric about $-d$, so the abscissa arm costs $\Theta(N)$ (measured $\frac{A}{N} = 0.236$). A post-hoc random-relation walk removes this.

1.3. psi versus one addition (measured, ns per operation, this host)

Pair 46: addition 462.4 ns, psi x-evaluation 454.2 ns, so $c_E = 0.982$. Legendre symbol: 497.2 ns. Toys: c_E = 1.06–1.14. Both operations are dominated by one inversion. In field operations, an affine addition is 1I+2M+1S and psi is 1I+10M. With batched inversion psi would cost about 2.2 additions (derived), so the measured $c_E \approx 1$ favours endo-chain.

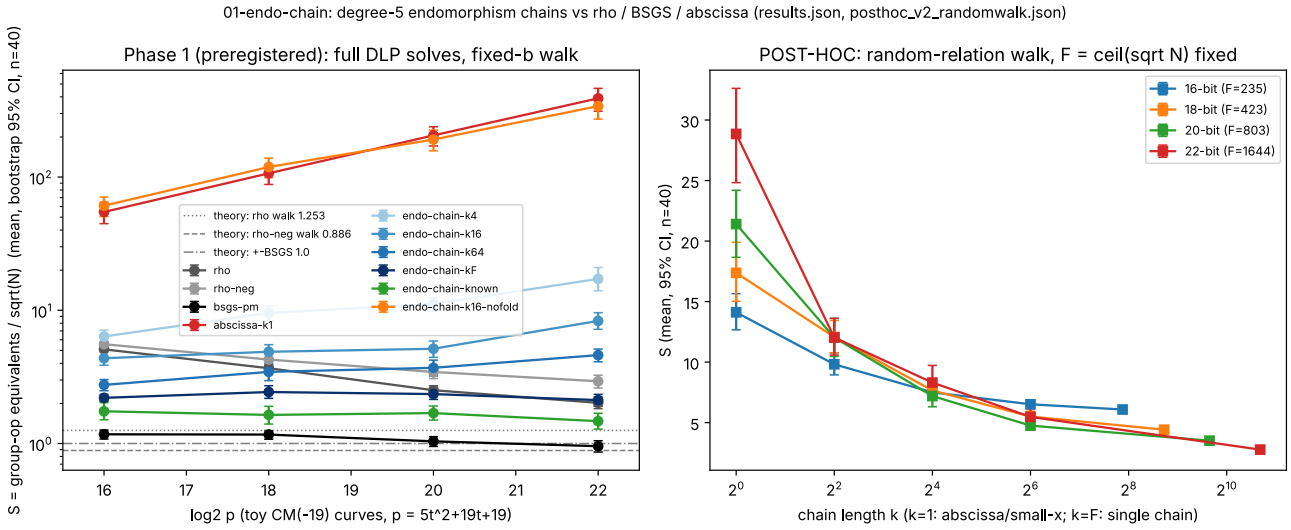


Figure 1: Left: phase 1 (preregistered), mean S with bootstrap 95% CI, $n = 40$ per arm and size (`results.json`). Right: post-hoc random-relation walk, S against chain length k at $F = \lceil \sqrt{N} \rceil$ (`posthoc_v2_randomwalk.json`).

1.4. Phase 1 headline values (S, mean [95% CI], 22-bit toy, N = 2699659)

rho (set-up included; walk alone 1.25)	2.02 [1.83, 2.24]
rho-neg (restarts included; walk alone 0.88)	2.93 [2.61, 3.26]
bsgs-pm	0.955 [0.86, 1.05]
abscissa-k1 (E5 artefact)	390 [312, 463]
endo-chain-k16	8.32 [7.20, 9.59]
endo-chain-kF (unknown head)	2.12 [1.90, 2.34]

endo-chain-known	1.47 [1.28, 1.69]
k16-nofold (positive control)	341 [272, 407]

Endo-chain-known / BSGS by size: 1.49, 1.41, 1.63, 1.54, all with CI above 1.18. The E3 prediction is $\sqrt{2c_E} \approx 1.46$–1.51. The fold's positive control (nofold / k16) ranges from 14 to 41, so the instrument sees the fold. Post-hoc ratio to abscissa at 22 bits: k16 0.29 [0.23, 0.36]; kF 0.096 [0.083, 0.114].

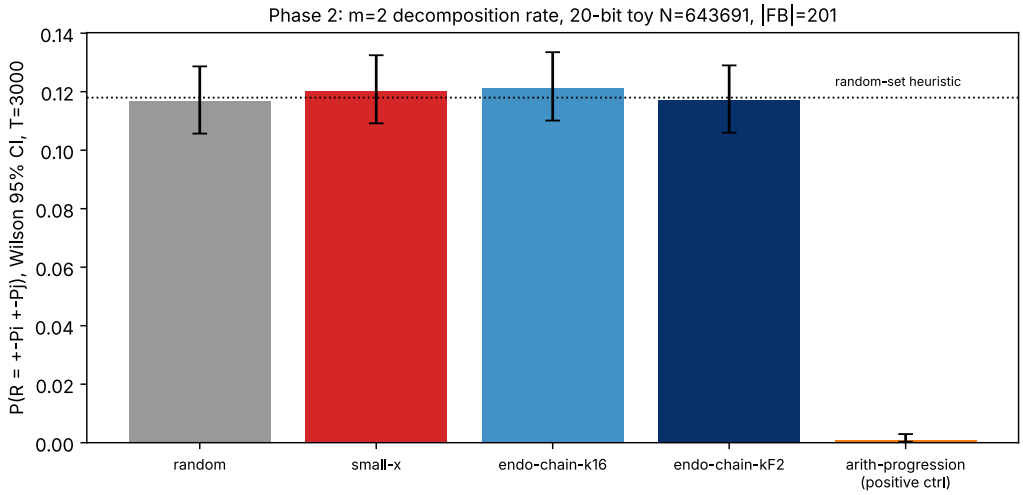


Figure 2: Phase 2 (preregistered): probability that a random point is $\pm P_i \pm P_j$, Wilson 95%, $T = 3000$, 20-bit toy, $|\mathrm{FB}| = 201$. The endo-chain sets fall inside the random-set null. The arithmetic-progression positive control is about 100 times lower.

1.5. Limits

Toy sizes only (16–22 bits; pair 46 was used for E1 and the timing). The host is shared, so timings are not isolated. Decomposition was tested for $m = 2$ only. This is not an ecbench session. Nothing was cut from the preregistered design, and the post-hoc arms are labelled. Reproduce with the commands in `RESULTS.md`.